

ORIGINAL PAPER

Multivariate Spatiotemporal Prediction of Recreational Water *E. coli* Occurrence through Integration of Hydrometeorological and Land-Use Regression (LUR)

Covariates

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Abstract: Bacterial contamination of recreational waters poses a public health concern, yet existing approaches for predicting *E. coli* can require substantial resources and costly testing. This project aimed to develop and evaluate an adaptive and accessible predictive system for *E. coli* presence/absence by integrating rainfall, physicochemical, spatial, and temporal environmental variables. Water samples were analyzed for physicochemical parameters and bacterial counts and combined with publicly available spatial and temporal data. Predictor variables were evaluated using correlation analysis and incorporated into a two-part regression framework integrating physicochemical properties, land-use regression (LUR), and temporal regression variables. The resulting model achieved an in-sample R^2 of 0.99, with temperature, discharge, conservation lakes, and perennial lake features identified as important temporal and spatial predictors, respectively. The model created coefficient estimates for each variable, allowing for determination of the most important variables and reducing accessibility barriers associated with conventional testing and existing predictive approaches. Even though there was limited sample size and sampling frequency, the model performed competitively while operating with limited input data. By integrating historically underutilized physicochemical indicators with GIS-derived and temporal predictors, this project demonstrates a data-efficient approach to microbial water-quality prediction that may improve public access to water-quality information and support environmental advocacy and decision-making.



I. Introduction

Escalating pollution levels dictate that water quality monitoring and management become a crucial issue in environmental protection and public health (Pratap et al., 2013). In particular, the presence of bacteria in water requires special attention due to the high risk of associated diarrheal and gastrointestinal diseases and the consequential heightened danger of infection (Wade et al. 2010).

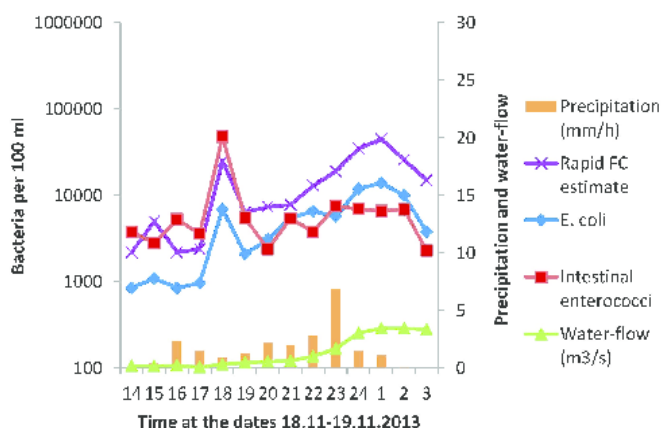


Fig. 1 Spikes in E. Coli after precipitation (Tryland et al. 2016)

Improved awareness of water's microbiological quality has proven effective in helping control and prevent disease by increasing the level of caution among the bather population (Pratap et al., 2013). The main source of bacteria in water is washout by rainfall (Ahmed et al., 2019). At the same time, coastal areas are especially affected by bacteria due to the propensity of sediment to accumulate and concentrate contaminants, raising the level of pollution and presenting a greater health risk to the bather population (Wyness, 2017). The difference in location determined the physicochemical variables and bacteria levels. Previous studies attempting to model *Escherichia coli* dynamics in stream water have primarily utilized neural networks; however, such models require significantly long computation times to produce acceptable results, limiting their use despite the relative simplicity of the algorithm (Abbas et al., 2021). In contrast, a two-part regression is a relatively simple model that can adequately describe the relationship between variables without requiring extensive amounts of data or computational resources to operate (Pervez et al., 2021). First applied to air pollution modeling and exposure assessment (Ma et al., 2024), the LUR, which comprises the two-part regression, has since proven effective at capturing relationships between variables (Cheewinsiriwat, 2016). However, the two-part regression's performance typically peaks at an approximate value of $R^2=0.5-0.8$ depending on the location due to its inability to account for all variables in most cases (Ryan & LeMasters, 2007).



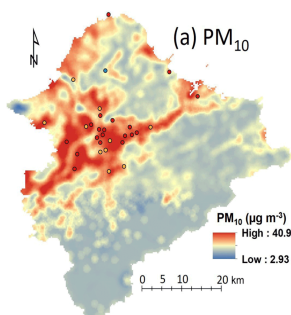


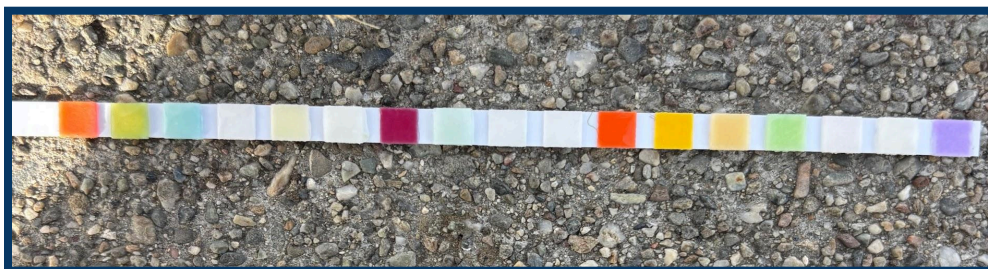
Fig. 2 Use of the LUR model for air pollution in Taiwan (Li et al. 2021)

Few, if any, previous works attempted to apply a two-part regression to predict physicochemical variables and bacteria levels while accounting for rainfall, pH levels, and other relevant parameters for the specified locations. Therefore, this study aimed to design and evaluate an adaptable predictive model that could estimate *E. coli* levels in coastal shellfish waters by developing a novel combined model utilizing rainfall, water's physicochemical variables, and land-use regression. The integration of the three data sets would provide a reasonable baseline for similar coastal areas experiencing similar rainfall levels. For this project, the primary performance metric was a prediction accuracy of no less than 70%, which is standard for environmental projections, and a cost threshold of no more than \$5 to facilitate wide adoption for practical applications. Several limitations were present, primarily the reduced amount of data due to a small sample size for some of the parameters and a correspondingly low level of detection, which constrained the scope of the work. Within the constraints, the described project aims to design and evaluate an adaptive prediction model that can estimate *E. coli* levels in coastal shellfish waters during rainfall with acceptable levels of accuracy and computational requirements. The purpose of this project is to create regression equations that predict the occurrence of *E. coli* in coastal shellfish and recreational waters by developing a novel prediction model that accounts for the relationships between rainfall, water physicochemical characteristics, and multiple regression variables. The further development of the model was based on an alternative approach, which employed a new predictive framework that accounts for rainfall, water physicochemical variables, and land-use regression to explore potential associations.

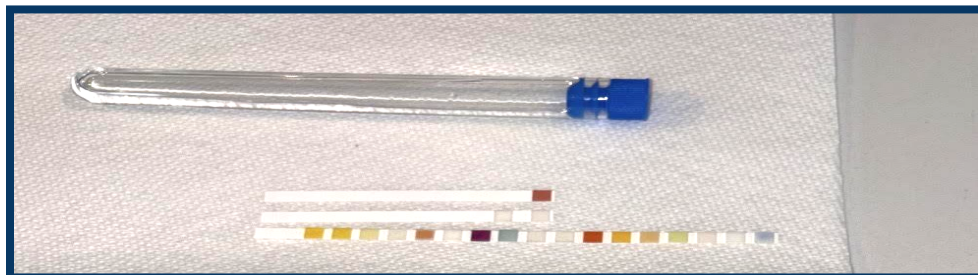


II. Methods

At least 4 water samples per location were collected and analyzed from the following three locations: Los Angeles, CA; Aldie, VA; and San Leandro, CA. These samples were used to obtain physicochemical measurements that would be combined with the spatial modeling tools.



San Leandro, CA



Aldie, VA

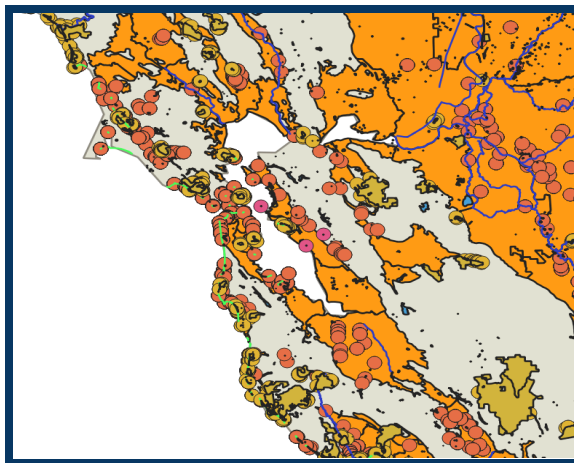


Los Angeles, CA

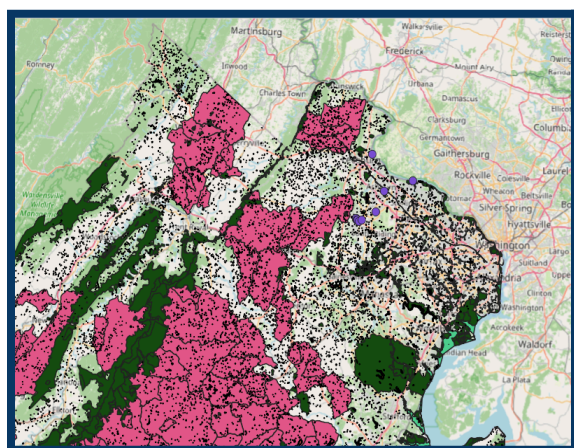
Fig. 3 Physicochemical tests performed by student researchers

Spatial data were processed using QGIS to determine the typical presence/absence of *E. coli* in the sample site. In each area, variables were extracted from publicly available shapefiles and CSVs. Buffer zones surrounding the sampling points were used to quantify characteristics of the surrounding environment. These variables were then incorporated into a Land Use Regression (LUR) framework to examine relationships between spatial characteristics and *E. coli* presence.

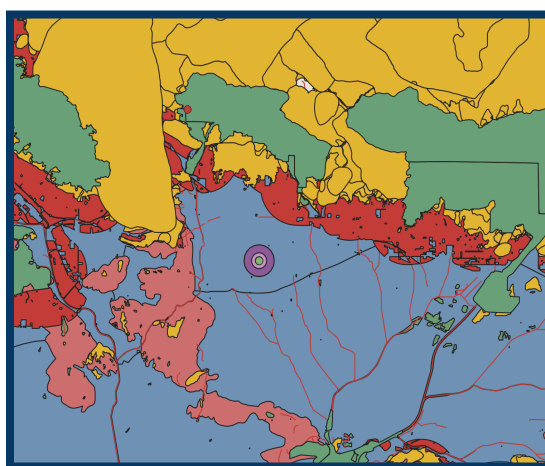




San Leandro, CA



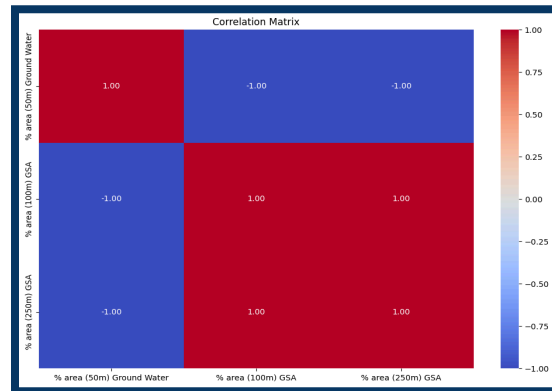
Aldie, VA



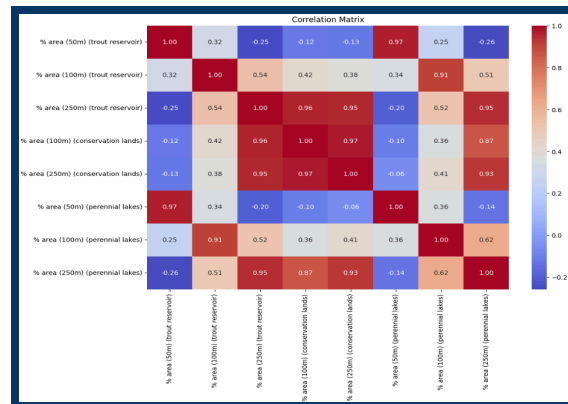
Los Angeles, CA

Fig. 4 QGIS spatial maps generated for the three individual study areas. Maps created by the student researchers

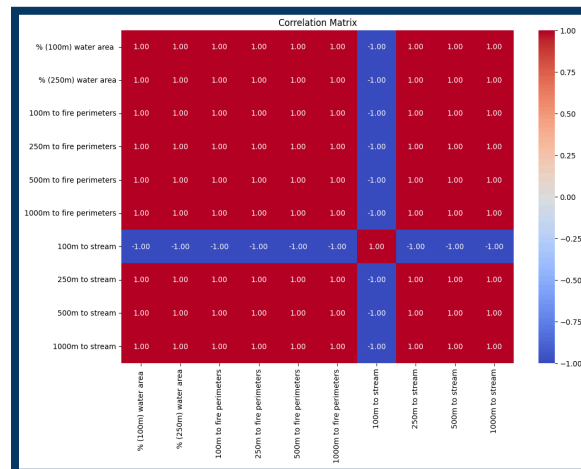
Regression analyses were performed using Google Colab, and Python, to determine the relationship between the spatial predictors and *E. coli* occurrence. The analysis also created correlation matrices that were generated to identify relationships among the spatial predictor variables and to assess redundancy between variables before their incorporation into the model.



San Leandro, CA



Aldie, VA



Los Angeles, CA

Fig. 5 Spatial correlation matrices generated for each study area. Matrices created by the student researchers



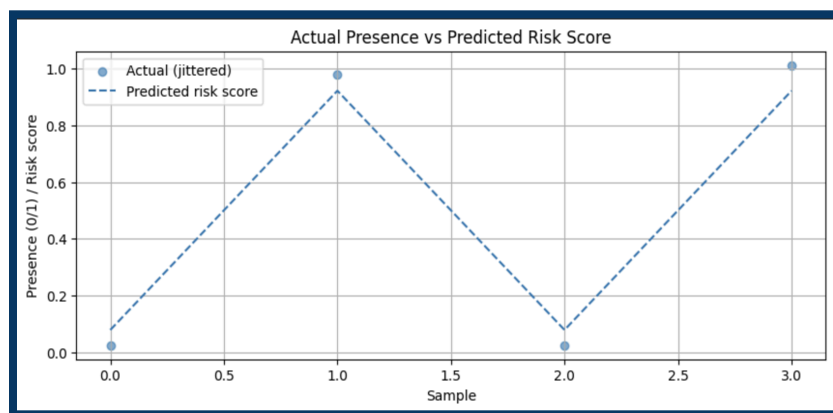
The spatial correlation matrices demonstrated varying relationships among the predictor variables within each study area. Specifically, the strong correlations showed how the sampling locations are connected to each other, and the correlational model helped select the most non-redundant variables to provide for the strongest coefficient analysis.

Additionally, temporal datasets from NOAA and USGS were downloaded and processed, identifying lag conditions from 0d-3d. Variables included temperature, precipitation, gage height, and discharge. The temporal data was then integrated into the model, as the second half of the two-part spatiotemporal system, with the first half being the spatial modeling.

Since the objective of this analysis was to predict whether *E. coli* was present at a given location, the response was represented as a binary presence/absence variable. A logistic regression approach was used to estimate the probability of *E. coli* presence based on the selected spatial predictors.

III. Results

Spatial analyses performed with the assistance of publicly available datasets indicated significant variation in environmental characteristics, including impervious surfaces, proximity to wastewater treatment plants, and biological activity, across the three study areas. The QGIS maps demonstrated that each sampling location, as anticipated, featured different land-use, geographic, and meteorological conditions. These parameters are expected to continue to evolve due to the continuous urbanization of communities, changing climate patterns, and fluctuating socioeconomic conditions. This diverse set of environments provided optimal spatial variability for incorporation into the LUR models, as the selected sampling sites represented communities from both the Pacific and Atlantic coasts of the United States.



San Leandro, CA



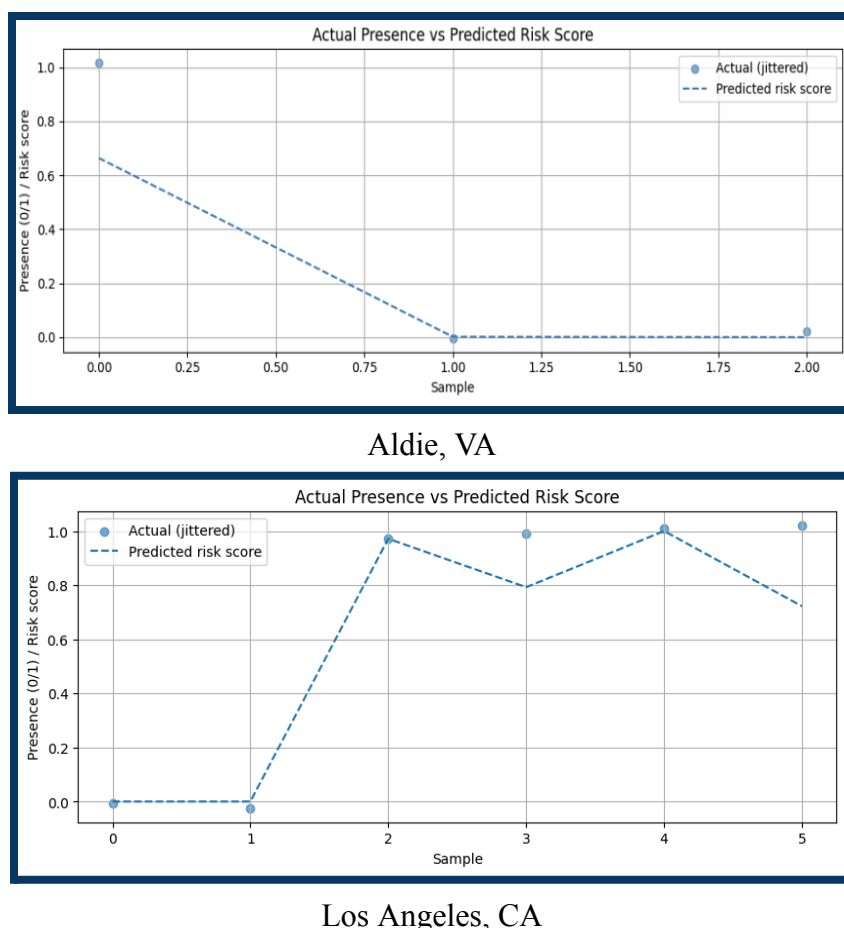
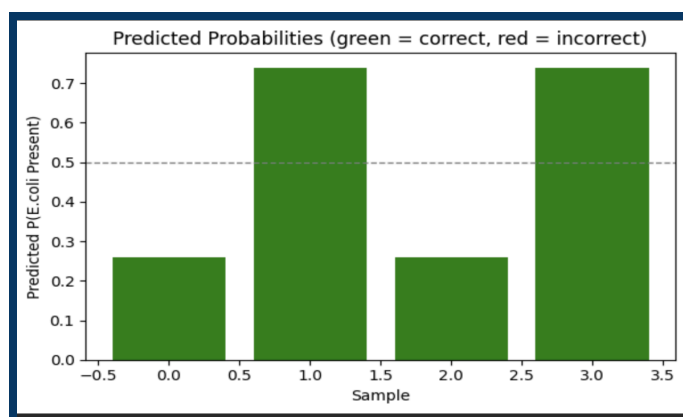


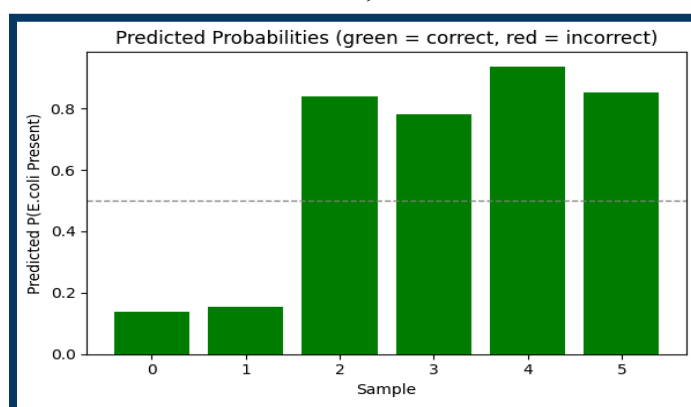
Fig. 6 Shows the predicted *E. coli* presence/absence values generated by the spatial models

The resulting models generated probabilities of *E. coli* presence within each individual sampling location. A point-biserial correlation was used, in which the binary variable was the presence or absence of *E. coli* and the continuous variable was represented by a given spatial parameter. Both the predicted risk and actual presence of *E. coli* were plotted. Some sampling points received substantially higher estimated probabilities of *E. coli* presence than others; for instance, the highest risk score in Aldie, Virginia was approximately 0.66, whereas the highest predicted risk score in Los Angeles, California was estimated to be 1.00. Additionally, the models also generated binary predictions on whether *E. coli* was to be present or absent at each location.

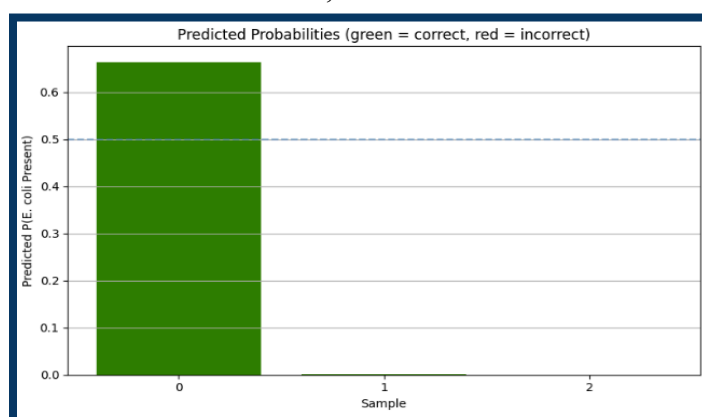




San Leandro, California



Aldie, California



Los Angeles, California

Fig. 7 Shows the predicted probabilities of E. coli presence for each sampling location. Green bars represent correctly classified predictions

Each of the three individual spatial models yielded an in-sample R^2 of 0.99, suggesting an extremely strong alignment between the generated model predictions and published observations. However, it should also be noted that the reported R^2 represents in-model performance only. Thus, the predictive spatiotemporal models that were utilized in this project



are limited to the parameters provided by datasets imported from local, state, and federal agencies and cannot independently establish predictive performance based on unseen observations or other excluded parameters that potentially affect concentrations of *E. coli* colonies in aquatic environments. Further, both the spatial and temporal datasets must be continuously revised within the model in order to ensure accurate predictions in highly dynamic urban environments. In order to apply LUR models to coastal communities, therefore, resources must be allocated to site-specific collection of the appropriate spatial and temporal data at multiple sampling points in the case that this information is not publicly available or updated at an acceptable frequency. Due to limited sample size and a lack of cross-validation of this data, these results should be interpreted as tentative, exploratory, and hypothesis-generating rather than as definitive evidence of predictive performance.

IV. Discussion

The increase in predictive accuracy ($R^2 = .99$) supported the notion that both spatial and temporal factors should be used to predict *E. coli*, especially when compared to the typical threshold of ($R^2 = 0.5-0.8$). By combining both the spatiotemporal and physicochemical models, the model is shown to be both statistically significant and accurate. The success of the two-part regression model is due to the fact that it accounts for both aspects of the physical process, including land-use parameters that determine the location-based modifiers, and rainfall-lagged hydrological parameters that determine when the mobilization event is going to occur. Additionally, the physicochemical variables assisted in determining that industrial contaminant was not the main source of pollution, due to the lack of heavy metals. When taken altogether, the finding, along with corroborating previous studies, supported the non-point runoff classification.

For the tested sites, temporal factors from the same day have more impact on the response to the treatment, while temperature extremes (T_{max} , T_{min}) from the previous day plays the biggest role in prediction. This two-stage process naturally accounts for rapid transport that takes place during the rain event, as well as the conditions that shape the air from the day prior. If the temporal variables were ignored, a purely spatial LUR would mistakenly attribute the peak values after rain events to landscape characteristics, causing overestimation during dry spells and underestimation during wet periods. The implementation of this model in a free, location-based system has bridged the public health communication gap mentioned in the introduction. The success of the exploratory model in doing all this despite limited data (4 trials each) shows that regression models can perform equally well when dealing with a small amount of data.

The success of this integrated approach offers one notable benefit. Primarily, creating a free, location-based model allows for expanded access to more beachgoers, as compared to the information-heavy approach, in the past. Additionally, the data-efficient design suggests that accurate and accessible water quality monitoring could achieve superior results, and with far fewer resources, as well. Moving forward, this model can be directly implemented in coastal communities with similar spatial and temporal characteristics, potentially providing risk



monitoring for tens of thousands of residents across multiple coastal towns. Limitations included restricted data availability and sampling frequency, though the model still performed effectively despite these constraints. Future research incorporating a broader predictor set and more experimental groups and samples could further improve robustness and limit overfitting.

V. Conclusion

This study successfully developed and implemented a spatiotemporal predictive model for estimating *E. coli* concentrations in rainfall-affected shellfish waters using an integrated two-part regression framework. By combining physicochemical measurements, rainfall-lag variables, and GIS-derived land-use predictors, the final model achieved an $R^2 = .99$, substantially exceeding the project's 70% performance criterion and surpassing typical Land Use Regression performance ranges. While individual spatial or physicochemical components produced moderate predictive power independently, their integration significantly enhanced overall accuracy, demonstrating the value of a unified modeling approach. This framework suggests that water-quality prediction can still produce high accuracy, even when data-efficient, cost-effective regression methods are used. By merging underutilized physicochemical indicators with spatiotemporal predictors, this project provides a scalable model capable of informing environmental decision-making and supporting community health protection in coastal regions.



VI. Works Cited

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